

# PERFORMANCE TESTING PHASE

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## Electric Vehicle Analytics Dashboard using Flask & Tableau E-CarStart

### Project Information

|               |                                      |
|---------------|--------------------------------------|
| Project       | Electric Vehicle Analytics Dashboard |
| Testing Type  | Functional & Performance Testing     |
| Platform      | Flask Web Application                |
| Visualization | Tableau Public                       |
| Browser       | Google Chrome / Microsoft Edge       |

### Introduction

This phase evaluates the functionality, responsiveness, reliability, and overall performance of the Electric Vehicle Analytics Dashboard. Testing ensures that every module performs as expected and provides a seamless user experience.

### Test Plan

Testing was performed on all website modules including Home, Dashboard, Story, and About Project pages. Navigation, embedded Tableau visualizations, responsiveness, and loading performance were verified.

### Functional Test Cases

| Test Case      | Expected Result      | Actual Result | Status |
|----------------|----------------------|---------------|--------|
| Open Home Page | Loads successfully   | Passed        | Pass   |
| Open Dashboard | Dashboard displayed  | Passed        | Pass   |
| Open Story     | Story displayed      | Passed        | Pass   |
| Navigation     | Links work correctly | Passed        | Pass   |

Responsive UI

Layout adapts

Passed

Pass

## Performance Testing

The application was tested for page loading, dashboard rendering, and navigation responsiveness. Tableau dashboards loaded successfully under a stable internet connection.

## Browser Compatibility Testing

The application was verified on Google Chrome and Microsoft Edge. All pages rendered correctly without layout issues.

## Responsive Testing

The website was tested on desktop and mobile screen sizes using Bootstrap's responsive layout. Navigation and content adjusted appropriately.

## User Acceptance Testing (UAT)

Users were able to navigate the application, interact with the Tableau dashboard and story, and understand EV analytics without functional issues.

## Test Results

All major functionalities passed testing. Dashboard embedding, story visualization, routing, and responsive design worked as expected.

## Bug Summary

Minor UI alignment and image path issues identified during development were resolved before final testing.

## Conclusion

Performance testing confirmed that the Electric Vehicle Analytics Dashboard is stable, responsive, and suitable for demonstrating EV insights through an interactive Flask and Tableau application.